SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 –Scenario Based Questions**	**Assessment:**	**Assessment Tool 1– Poster Presentation**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B. Ganga Jayadev Reddy	**Reg. No.:**	192472248
Section / Batch:	CSE AI	**Date:**	21 - 08 - 2026

Code of Conduct

I B. Ganga Jayadev Reddy (192472248)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Ganga Jayadev Reddy

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Hypothesis Testing Using Confidence Interval

Question:
A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Answer:
The university claims that at least 80% of its students are satisfied with online classes. This means the population satisfaction proportion is expected to be 80% or more.

The 95% confidence interval obtained from the survey is:
Confidence Interval = (74%, 79%)

The claimed value of 80% does not fall within this confidence interval. In fact, the entire confidence interval ranges from 74% to 79%, which is below the claimed value of 80%.

We can express the hypotheses as:
H_0: $p \geq 0.80$
H_1: $p < 0.80$

Since the confidence interval does not contain 80%, the survey results provide evidence against the university's claim.

Conclusion:
At the 95% confidence level, the university's claim that at least 80% of students are satisfied with online classes is not supported by the survey data. The results suggest that the actual satisfaction proportion is likely to be less than 80%.

2. Hypothesis Testing Using Confidence Interval

Question:
A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Answer:
The developer claims that users spend an average of 30 minutes per day on the application.

The hypotheses can be stated as:
H_0: $\mu = 30$ minutes
H_1: $\mu \neq 30$ minutes

The sample produces a 99% confidence interval of:
(31.5, 35.2) minutes

For the claim to be supported, the claimed population mean of 30 minutes should fall within the confidence interval.

However, 30 minutes is outside the confidence interval, and the entire interval is greater than 30 minutes.

This indicates that the sample data provides strong evidence that the actual average time spent by users is different from 30 minutes. Since the whole interval is above 30 minutes, it specifically suggests that the actual average is greater than 30 minutes.

Conclusion:
The developer's claim that users spend an average of 30 minutes per day is not supported by the sample data. At the 99% confidence level, the actual average usage time is likely to be between 31.5 and 35.2 minutes per day, which is greater than the claimed 30 minutes.

3. Hypothesis Testing Using p-value

Question:
A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Answer:
The credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions.

The hypotheses can be written as:
H_0: $p = 0.95$
H_1: $p \neq 0.95$

The given values are:
p-value = 0.02
Significance level, $\alpha = 0.05$

The decision rule is:
If p-value $\leq \alpha$, reject H_0.
If p-value $> \alpha$, fail to reject H_0.

Here:
$0.02 < 0.05$

Therefore, the p-value is smaller than the significance level. Hence, we reject the null hypothesis.

This means that the observed sample results would be relatively unlikely if the actual fraud detection rate were exactly 95%.

Conclusion:
At the 5% significance level, there is sufficient statistical evidence to reject the company's claim that the fraud detection system identifies exactly 95% of fraudulent transactions. The data suggests that the actual detection rate is significantly different from 95%.

4. Hypothesis Testing Using p-value

Question:
An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Answer:
The existing recommendation algorithm has a conversion rate of 12%. After introducing the new algorithm, the observed conversion rate increases to 15%.

The hypotheses can be stated as:
H_0: The new algorithm does not improve the conversion rate.
H_1: The new algorithm improves the conversion rate.

The given information is:
Existing conversion rate = 12%
New observed conversion rate = 15%
p-value = 0.03
$\alpha = 0.05$

We compare the p-value with the significance level:
p-value = 0.03
$\alpha = 0.05$

Since:
$0.03 < 0.05$

we reject the null hypothesis.

The observed conversion rate has increased from 12% to 15%, and the p-value indicates that this improvement is statistically significant at the 5% significance level.

Therefore, the improvement is unlikely to be explained only by random sampling variation.

Conclusion:
At the 5% significance level, there is sufficient statistical evidence to conclude that the new recommendation algorithm significantly improves the conversion rate. The conversion rate increased from 12% to 15%, and the hypothesis test confirms that this improvement is statistically significant.

5. Hypothesis Testing Using p-value

Question:
A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Answer:
The hospital has introduced an AI-based patient scheduling system and wants to determine whether the new system reduces the average waiting time of patients.

The hypotheses can be stated as:
H_0: The AI-based scheduling system does not reduce the average waiting time.
H_1: The AI-based scheduling system reduces the average waiting time.

The given values are:
p-value = 0.08
Significance level, $\alpha = 0.05$

The decision rule is:
If p-value ≤ 0.05, reject H_0.
If p-value > 0.05, fail to reject H_0.

Now compare:
$0.08 > 0.05$

Since the p-value is greater than the significance level, we fail to reject the null hypothesis.

This means that the evidence from the sample is not strong enough to conclude that the AI-based scheduling system has significantly reduced the average waiting time.

It is important to note that failing to reject the null hypothesis does not prove that the system has no effect. It simply means that the available sample evidence is insufficient to establish a statistically significant reduction at the 5% level.

Conclusion:
At the 5% significance level, the hospital should not conclude that the AI-based scheduling system significantly reduces average waiting time. Since the p-value of 0.08 is greater than 0.05, there is insufficient statistical evidence to support the claim that the new system reduces waiting time.